Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

University of Pisa, a.y. 2025-2026

Alessandro Gori*

Thesis for the Master's degree in Physics

Abstract

[To be continued...]

Contents

1 Mean-field theory discussion of the EHM **1**
1.1 Extended Hubbard model symmetries and Wick's theorem . . . 1
1.1.1 Spatial and topological symmetries . . . 1
1.1.2 Spin and charge symmetries . . . 2
1.1.3 Particle-hole symmetry at half-filling . . . 3
1.2 MFT on Hubbard lattices . . . 5
1.2.1 MFT general procedure, self-consistency and Wick's theorem . . . 5
1.2.2 MFT treatment of the pure Hubbard model . . . 6
1.3 The non-local attraction $\hat{H}_V$ as a source of symmetry-breaking interactions . . . 7
1.3.1 Real space description . . . 7
1.3.2 Reciprocal space transform of the model interactions . . . 8
1.4 General approach MFT through iterative Hartree-Fock algorithms . . . 10
1.4.1 Sketch of the Hartree-Fock iterative algorithm . . . 10
1.4.2 Self-consistency and RMPs . . . 12
1.4.3 Fixed density simulations and the role of μ . . . 12
1.5 Square lattice spatial symmetry channels . . . 13

Draft: February 9, 2026

*a.gori23@studenti.unipi.it / nepero27178@github.com

Chapter 1

Mean-field theory discussion of the EHM

This chapter is devoted to develop a Mean Field Theory (MFT) framework of the Extended Hubbard model (EHM) of Eq. (**??**),

$$\hat{H} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma}}_{\hat{H}_t} \underbrace{+U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\hat{H}_U} \underbrace{-V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{n}_{i\sigma} \hat{n}_{j\sigma'}}_{\hat{H}_V}$$

Mean Field Theory (MFT) is a widely used and simple theoretical tool, often sufficient to describe the leading orders in phase transition phenomena of Many-Body Physics. Here MFT is employed generically, in order to later discuss both the effects of the non-local term $\hat{H}_V$ onto the AF phase (Chap. **??**), as well as the insurgence of anisotropic superconductivity (Chap. **??**). The latter, following the path of Bardeen-Cooper-Schrieffer (BCS) theory in describing conventional s-wave superconductivity. As will be thoroughly described, the lattice spatial structure directly influences the topology of the gap function, giving rise to anisotropic pairing.

1.1 Extended Hubbard model symmetries and Wick's theorem

The general aim of this project is to study the phase diagram of the EHM by comparing ground-state energies of different phases. The phases we consider in next chapters for the EHM are the normal state (N), the antiferromagnetic ordering (AF), given by a non-uniform distribution of charge in each spin sector, and the anisotropic superconducting (SC) phase, described by a uniformly distributed charge allowing for Cooper pairing instabilities.

At the end of this section, in Tab. 1.2 we summarize all symmetries with the relative operations. Throughout the entire thesis we will refer to specific SSB for given groups to establish a specific phase.

1.1.1 Spatial and topological symmetries

Let us start from the trivial topology of the lattice we are working on. The EHM of Eq. (**??**) is evidently translational invariant on both the x and y directions. We refer to these groups as $\mathrm{T}_1^{(x)}$, $\mathrm{T}_1^{(x)}$. The 1 subscript indicates the one-site translations under which the model is symmetric.

Evidently, the following statement holds:

$$\mathrm{T}_n^{(\ell)} \supset \mathrm{T}_{kn}^{(\ell)} \qquad k \in \mathbb{R}$$

A model which is invariant for n-sites translations in the ℓ direction, also is for kn-sites translations. Then we can break translation symmetry in a given direction reducing it to a smaller group, say

$$\mathrm{T}_1^{(\ell)} \quad \overset{\text{SSB}}{\rightarrow} \quad \mathrm{T}_2^{(\ell)}$$

This is precisely what happens in the AF phase.

Another natural symmetry of the EHM is the four-fold point group C_4 of $\pi/2$ rotations, expressing the square lattice property being mapped onto itself under said rotations. A similar property as for the translations hold for point groups,

$$\mathrm{C}_n \supset \mathrm{C}_{n/k} \qquad k, n/k \in \mathbb{R}$$

which expresse the fact that, if a system is topologically invariant under n-fold rotations and $k \in \mathbb{R}$ exists such that $n/k \in \mathbb{R}$, then the smaller point group is automatically included. As for translations, we can break the point group reducing it to a smaller subgroup, say

$$\mathrm{C}_4 \overset{\text{SSB}}{\rightarrow} \mathrm{C}_2$$

This is what happens with the effective arising of anisotropic corrections to the kinetic hamiltonian due to non-local interactions, as will be discussed in Sec. **??**.

1.1.2 Spin and charge symmetries

When it comes to Fermi algebra and operators, an evident symmetry of the EHM is a global U(2) symmetry [**arovas2022hubbard**] given by the transformation

$$\hat{c}_{i\sigma} \rightarrow \sum_{\sigma'} \mathcal{U}_{\sigma\sigma'} \hat{c}_{i\sigma'} \quad \text{with} \quad \mathcal{U}_{\sigma\sigma'} \in \mathrm{U}(2) \tag{1.1}$$

The proof is trivial and follows from the fact that $\mathcal{U}^\dagger \mathcal{U} = \mathcal{U}\mathcal{U}^\dagger = \mathbb{I}$. Note that the the following decomposition holds,

$$\mathrm{U}(2) = \mathrm{SU}(2) \otimes \mathrm{U}^{\mathrm{c}}(1) \tag{1.2}$$

which expresses the separate charge conservation and spin rotational symmetries. In detail:

- $\mathrm{U}^{\mathrm{c}}(1)$ charge conservation, coming from Eq. (1.2) decomposition. By breaking this symmetry, we allow for the total charge to fluctuate in the ground state. This fluctuation is only allowed in the SC phase in the present text;

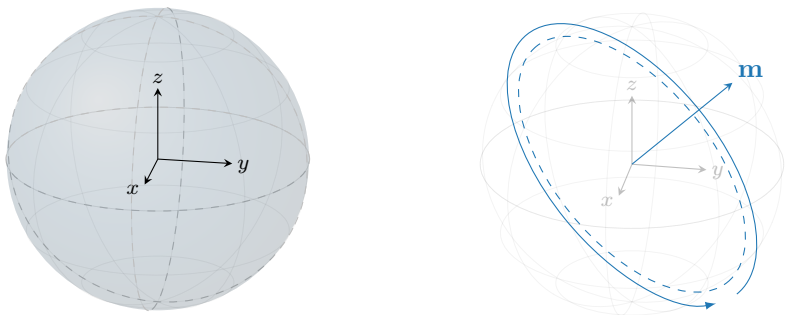
- SU(2) spin rotation symmetry, also , coming from Eq. (1.2) decomposition. Note that:

  $$\mathrm{SU}(2) \supset \mathrm{U}^z(1)$$

  The z subscript we included to distinguish from the charge group. Hence SU(2) symmetry can be broken, selecting a particular vectorial direction for the order parameter (we call z), and reduced to a smaller $\mathrm{U}^z(1)$ symmetry which is essentially expressed by the conservation of the magnitude of the order parameter.

The last point is important: for the AF phase, for example, there is no need for the magnetization vector to be directed along the physical z-axis, which is the one perpendicular to the lattice plane. SU(2) symmetry breaks when (staggered) magnetization is established along a particular direction, but spin rotations around said direction still are ground state symmetries. This is shown schematically in Fig. 1.1.

Different phases are described by different order parameters, each one exhibiting a specific symmetry subset from the initial set, while the rest are broken. Ferromagnetic state perform $\mathrm{SU}(2) \rightarrow \mathrm{U}^z(1)$ spontaneous symmetry breaking (SSB), through the magnetic order parameter acquiring a specific spatial direction. AF ordering breaks also translational invariance (reducing it to a smaller symmetry, relative to double-sized unitary cells). Basic SC models do not break translational invariance and can preserve full SU(2) symmetry, while breaking $\mathrm{U}^{\mathrm{c}}(1)$ charge conservation. None of the phases hereby considered do completely break translational invariance: the latter either is left untouched or is reduced to a weaker translational invariance. Because of this, it will be rather useful to move the framework to reciprocal space. We will get there in Sec. **??**. This discussion is summarized in Tab. 1.1, where we report the various SSBs for AF and SC phases.



a | SU(2) rotations domain before SSB.

b | U(1) rotations domain after SSB.

Figure 1.1 | Graphic representation of the concept of SU(2) → U(1) SSB. The blue color identifies, in both cases, the rotations domain for the current symmetry. When SSB takes place in a magnetic phase, the magnetization order parameter acquires a precise direction which was previously invariant over the full spherical domain. After SSB, the hamiltonian is rotationally invariant only for rotations *around* the selected magnetization vector.

Phase	**SSBs**
AF	$\mathrm{T}_1^{(\ell)} \to \mathrm{T}_2^{(\ell)}$
	$\mathrm{SU}(2) \to \mathrm{U}^z(1)$
SC	$\mathrm{U}^c(1) \to 0$

Table 1.1 | SSB interpretation of the AF and SC phases. In the last line, the 0 on the SSB right-hand side indicates a completely disrupted symmetry.

1.1.3 Particle-hole symmetry at half-filling

The half-filled EHM is of particular interest due to its particle-hole symmetry (PHS), which is immediately distrupted at infinitesimal doping. Let us prove that for the EHM. Consider the generic unitary operation

$$\mathrm{P} \quad : \quad \hat{c}_{i\sigma} \quad \to \quad e^{i\eta_{i\sigma}} \hat{h}^\dagger_{i\sigma}$$

being in general η_i a phase dependent on the fermionic quantum state $(i\sigma)$. Inserting the above transformation in the EHM, we get

$$\mathrm{P}\{\hat{H}\} = -t \sum_{\langle ij \rangle} \sum_{\sigma} e^{i(\eta_{i\sigma}+\eta_{j\sigma})} \hat{h}_{i\sigma} \hat{h}^\dagger_{j\sigma} + U \sum_i \hat{h}_{i\uparrow} \hat{h}_{i\downarrow} \hat{h}^\dagger_{i\downarrow} \hat{h}^\dagger_{i\uparrow} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}_{i\sigma} \hat{h}_{j\sigma'} \hat{h}^\dagger_{j\sigma'} \hat{h}^\dagger_{i\sigma}$$

For the two interactions $\hat{H}_U$, $\hat{H}_V$ the phase factors cancelled out. We aim to find the conditions under which the above hamiltonian is mapped onto the EHM of (**??**),

$$\mathrm{P}\{\hat{H}\} \stackrel{?}{=} -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{h}^\dagger_{i\sigma} \hat{h}_{j\sigma} + U \sum_i \hat{h}^\dagger_{i\uparrow} \hat{h}^\dagger_{i\downarrow} \hat{h}_{i\downarrow} \hat{h}_{i\uparrow} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}^\dagger_{i\sigma} \hat{h}_{j\sigma'} \hat{h}_{j\sigma'} \hat{h}_{i\sigma}$$

Before starting the derivation, let us discuss some properties of the hole operators. For the hole operators, due to the presence of the phase factors in the PHS operative definition, the Fermi anticommutation rules hold regardless of the phase factor:

$$\begin{aligned} \{\hat{h}_{i\sigma}, \hat{h}^\dagger_{j\sigma'}\} &= e^{i(\eta_i - \eta_j)} \{\hat{c}^\dagger_{i\sigma}, \hat{c}_{j\sigma'}\} \\ &= e^{i(\eta_i - \eta_j)} \delta_{ij} \delta_{\sigma\sigma'} \\ &= \delta_{ij} \delta_{\sigma\sigma'} \end{aligned}$$

and evidently:

$$\hat{h}_{i\sigma} \hat{h}^\dagger_{i\sigma} = \hat{n}_{i\sigma}$$

being $\hat{n}_{i\sigma}$ the fermionic number operator on site i with spin σ. It follows, for quartic interactions

$$\begin{aligned}
\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'}\hat{h}^\dagger_{i\sigma} &= -\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'} \\
&= \hat{h}_{i\sigma}\hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\
&= -\hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\
&= \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\
&= -\hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{i\sigma}\hat{h}_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \left(\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\right)\delta_{ij}\delta_{\sigma\sigma'} \\
&= \hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} + (\hat{n}_{i\sigma} + \hat{n}_{j\sigma'}) - 1 - \left(\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\right)\delta_{ij}\delta_{\sigma\sigma'}
\end{aligned}$$

The first term of the last passage is mirrored with respect to the starting term. Now, let us deal with the kinetic, local and non-local interactions separately.

Kinetic term. Consider the kinetic term,

$$\mathrm{P}\{\hat{H}_t\} = -t\sum_{\langle ij\rangle}\sum_{\sigma} e^{i(\eta_{i\sigma}+\eta_{j\sigma})}\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma}$$

It suffices to anticommute the two fermionic operators, and exchange site labels $i \leftrightarrow j$, to get

$$\mathrm{P}\{\hat{H}_t\} = -t\sum_{\langle ij\rangle}\sum_{\sigma} e^{i(\eta_{i\sigma}+\eta_{j\sigma}+\pi)}\hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma}$$

A necessary condition on the unitary operation P is the following,

$$\eta_{i\sigma} + \eta_{j\sigma} = (2k-1)\pi$$

valid for NN sites.

Local repulsion. Substitute in $\hat{H}_U$ the above derivation,

$$\begin{aligned}
\mathrm{P}\{\hat{H}_U\} &= U\sum_i \hat{h}_{i\uparrow}\hat{h}_{i\downarrow}\hat{h}^\dagger_{i\downarrow}\hat{h}^\dagger_{i\uparrow} \\
&= U\sum_i \hat{h}^\dagger_{i\uparrow}\hat{h}^\dagger_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} + U\sum_i(\hat{n}_{i\uparrow} + \hat{n}_{i\downarrow}) - U\sum_i(1) \\
&= U\sum_i \hat{h}^\dagger_{i\uparrow}\hat{h}^\dagger_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} + U(\hat{N} - L_xL_y)
\end{aligned}$$

having we used in the last passage the inverse $\hat{h} \to \hat{c}$ map to recognize a number operator. Thus, a state realizing

$$\langle\hat{N}\rangle = L_xL_y$$

preserves PHS in the local sector.

Non-local attraction. Proceed identicall for $\hat{H}_V$,

$$\begin{aligned}
\mathrm{P}\{\hat{H}_V\} &= -V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'}\hat{h}^\dagger_{i\sigma} \\
&= -V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} - V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}(\hat{n}_{i\sigma} + \hat{n}_{j\sigma'}) + V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}(1) \\
&= -V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} + 2zV(\hat{N} - L_xL_y)
\end{aligned}$$

with $z = 4$ the lattice coordination number. As before, an half-filled phase preserves PHS in the non-local sector. In last passage we used the fact that summing over NNs means overcounting each site $z/2$ times, and spin degeneracy.

Symmetry	Operations
T_ℓ	Discrete translations in the $\ell \in \{x, y\}$ directions separately
C_4	Four-fold point group
$\mathrm{U}^{\mathrm{c}}(1)$	Charge-conserving global phase shifts
SU(2)	Three-dimensional rotations in spin space
PHS	Particle-hole exchange (only at half filling)
$\mathrm{U}^{z}(1)$	Reduced two-dimensional rotations around order parameter after SSB

Table 1.2 Model symmetries and relative group operations. Note that $\mathrm{U}^{\mathrm{c}}(1)$ represents the charge conserving group, while $\mathrm{U}^{z}(1)$ represents the subgroup obtained by breaking the SU(2) symmetry with a vector order parameter and preserving its amplitude. Since the latter is "derived" from SU(2), we write it separated from the four main symmetries of the model.

No other condition on phases is derived, thus we conclude the half-filled EHM preserves PHS if the two sublattices $\mathcal{S}_a$, $\mathcal{S}_b$ the square lattice can be divided into, being a (trivial) bipartite lattice, differ by a π phase difference under PHS transform. In other words, if P is defined by the unitary operation

$$\mathrm{P} \quad : \quad \begin{cases} \hat{c}_{i\sigma} \to \hat{h}^\dagger_{i\sigma} & \text{for} \quad i \in \mathcal{S}_a \\ \hat{c}_{i\sigma} \to -\hat{h}^\dagger_{i\sigma} & \text{for} \quad i \in \mathcal{S}_b \end{cases}$$

and the system is half-filled, PHS is respected. Any other phase choice, as long as it alternates by π on the two sublattices, is good enough. To summarize, we report in Tab. 1.2 all identified symmetries.

1.2 MFT on Hubbard lattices

This section serves as groundwork for everything that follows. We here define the MFT scheme as well as the iterative HF computational scheme we use to estimate parametrically the relevant physical quantities of the various phases. We start by recalling the general features of MFT and then move to the specificity of the EHM, apply Wick's theorem and finally move to algorithic aspects.

1.2.1 MFT general procedure, self-consistency and Wick's theorem

The Mean Field Theory approach to any fermionic quartic operator is given by the following procedure: given the hamiltonian made of the quadratic operators $\hat{A}$ and $\hat{B}$

$$\hat{H} \equiv \hat{A}\hat{B}$$

we define the fluctuations

$$\delta\hat{A} \equiv \hat{A} - \langle\hat{A}\rangle \qquad \delta\hat{B} \equiv \hat{B} - \langle\hat{B}\rangle$$

thus the following chain holds:

$$\begin{aligned} \hat{A}\hat{B} &= \langle\hat{A}\rangle\langle\hat{B}\rangle + \delta\hat{A}\langle\hat{B}\rangle + \langle\hat{A}\rangle\delta\hat{B} + \delta\hat{A}\delta\hat{B} \\ &= -\langle\hat{A}\rangle\langle\hat{B}\rangle + \hat{A}\langle\hat{B}\rangle + \langle\hat{A}\rangle\hat{B} + \underbrace{\delta\hat{A}\delta\hat{B}}_{\mathcal{O}(\delta)^2} \end{aligned}$$

which implies

$$\langle\hat{H}\rangle = \langle\hat{A}\rangle\langle\hat{B}\rangle + \langle\delta\hat{A}\delta\hat{B}\rangle$$

Thus, if $|\langle\hat{A}\rangle\langle\hat{B}\rangle| \gg |\langle\delta\hat{A}\delta\hat{B}\rangle|$, in a statistical sense we can neglect the $\mathcal{O}(\delta)^2$ part of the hamiltonian and approximate

$$\hat{H} \simeq \hat{A}\langle\hat{B}\rangle + \langle\hat{A}\rangle\hat{B} - \langle\hat{A}\rangle\langle\hat{B}\rangle \qquad \Longrightarrow \qquad \langle\hat{H}\rangle \simeq \langle\hat{A}\rangle\langle\hat{B}\rangle \tag{1.3}$$

Term	Normal phase	Antiferromagnet	Superconductor
Hartree			
Fock	Prohibited	Prohibited	Prohibited
Cooper	Prohibited	Prohibited	

Table 1.3 Summary of Wick's terms selection rules and relevant contributions to the effective hamiltonian for each phase limitedly to $\hat{H}_U$.

Self consistency of MFT procedure is thus embodied by the condition

$$\left|\frac{\langle\delta\hat{A}\delta\hat{B}\rangle}{\langle\hat{A}\rangle\langle\hat{B}\rangle}\right| \ll 1 \quad \Longrightarrow \quad \left|\frac{\langle\hat{A}\hat{B}\rangle}{\langle\hat{A}\rangle\langle\hat{B}\rangle} - 1\right| \ll 1 \tag{1.4}$$

which essentially states that the least the fluctuations, the better the approximation. This rule of thumb is necessary in order to interpret the physical relevance of numerical MFT results. We will come back to this in the end, for now assuming the validity of MFT approximation and deriving the mathematical results.
[To be continued...]

1.2.2 MFT treatment of the pure Hubbard model

The pure Hubbard model, $\hat{H}_t + \hat{H}_U$, offers a simple framework for MFT analysis of the AF phase: this is described in detail in App. **??**. The key passage is there given by the MFT approximation

$$\hat{n}_{i\uparrow}\hat{n}_{i\downarrow} \simeq \hat{n}_{i\uparrow}\langle\hat{n}_{i\downarrow}\rangle + \langle\hat{n}_{i\uparrow}\rangle\hat{n}_{i\downarrow} + (\text{constants}) \tag{1.5}$$

from which the AF structure is simply recovered. In terms of the discussion of the above paragraph, we have $\hat{A} \equiv \hat{n}_{i\uparrow}$ and $\hat{B} \equiv \hat{n}_{i\downarrow}$. However the "full" operator is given by

$$\hat{n}_{i\uparrow}\hat{n}_{i\downarrow} = \hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\downarrow}\hat{c}_{i\uparrow}$$

which contracts in three different ways following Wick's Theorem:

$$\langle\hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow}\hat{c}_{i\uparrow}\rangle \simeq \underbrace{\langle\hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i\downarrow}\rangle\langle\hat{c}_{i\downarrow}\hat{c}_{i\uparrow}\rangle}_{\text{Cooper}} - \underbrace{\langle\hat{c}^\dagger_{i\uparrow}\hat{c}_{i\downarrow}\rangle\langle\hat{c}^\dagger_{i\downarrow}\hat{c}_{i\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle\hat{c}^\dagger_{i\uparrow}\hat{c}_{i\uparrow}\rangle\langle\hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow}\rangle}_{\text{Hartree}} \tag{1.6}$$

As a first approximation, the theorem is assumed to hold (which, in a BCS-like fashion, is equivalent to assuming for the ground-state to be a coherent state). The AF ground state reduces translational invariance and rotational symmetry, as explained in Tab. 1.1. Thus, of the three terms above, when considering the pure Hubbard model in AF phase:

- Cooper fluctuations are suppressed, because they break charge conservation;
- Similarly the Fock term is null as well because if $i = j$ and $\sigma' = \overline{\sigma}$ (as is for the local interaction, which contains the operator $\hat{n}_{i\uparrow}\hat{n}_{i\downarrow}$) the expectation values involved are describing a process breaking the survivor $\mathrm{U}^z(1)$ spin symmetry.

Thus, correctly, the Wick's decomposition of pure HM only involves Hartree-terms of Eq. (1.6), getting us back to (1.5). The distinction between the normal phase and the AF phase id given by whether the Hartree fluctuations break or not translational symmetry: antiferromagnetism is indeed a specific SDW ordering which weakens the latter. In general the three parts of the decomposition (Hartree, Fock, Cooper) need to be considered altogether: this is what is done in the next chapters. What said here extends naturally to the EHM: the relevant contributions for each phase given by the local repulsion $\hat{U}$ in the EHM are listed in Tab. 1.3.

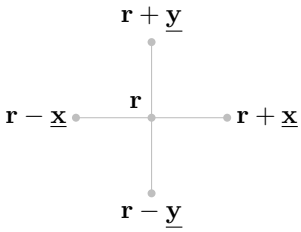


Figure 1.2 | Schematic representation of the four NNs of a given site i for a planar square lattice.

1.3 The non-local attraction $\hat{H}_V$ as a source of symmetry-breaking interactions

Let us now extend the discussion to the EHM and consider the NN non-local attraction,

$$\hat{H}_V \equiv -V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma\sigma'} \hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}'\sigma'} \tag{1.7}$$

where we changed notation, and replaced the site index i with its 2D position on the lattice

$$i \quad \rightarrow \quad \mathbf{r} = (x, y)$$

Let also the lattice versors,

$$\underline{\mathbf{x}} \equiv (1, 0) \qquad \underline{\mathbf{y}} \equiv (0, 1)$$

as in Fig. 1.2.

Evidently the hamiltonian can be decomposed in various spin terms,

$$\begin{aligned} \hat{H}_V &= \sum_{\sigma\sigma'} \hat{H}_V^{\sigma\sigma'} \\ &= \underbrace{\hat{H}_V^{\uparrow\uparrow} + \hat{H}_V^{\downarrow\downarrow}}_{\text{Same-spin}} + \underbrace{\hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow}}_{\text{Opposite-spin}} \end{aligned} \tag{1.8}$$

The role of said terms is crucial in establishing the pairing channel of the dominant physical phase. As an example, the s.s. terms can be seen as a possible source of triplet pairing superconductivity. Next section are devoted to analyze said terms and derive a simple analytical result in reciprocal space.

1.3.1 Real space description

Let us start from the real space form of Eq. (1.8). To carry out a summation over nearest neighbors $\langle \mathbf{r}\mathbf{r}' \rangle$ of a square lattice means precisely to sum over all links of the lattice. Then we can identify the generic opposite-spin (o.s.) term $\hat{H}_V^{\sigma\overline{\sigma}}$ as the one collecting the σ operators of sublattice $\mathcal{S}_a$ and $\overline{\sigma}$ operators of sublattice $\mathcal{S}_b$. The o.s. non-local interactions can be written as a sum of terms over just one of the two sublattices $\mathcal{S}_a$ and $\mathcal{S}_b$. Define the site hamiltonian

$$\hat{h}_V^{(\mathbf{r})} \equiv -V \sum_{\boldsymbol{\delta}} \left(\hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow} + \hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}-\boldsymbol{\delta}\downarrow} \right) \quad \text{with} \quad \boldsymbol{\delta} \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\} \tag{1.9}$$

Then, it follows,

$$\begin{aligned} \hat{H}_V^{(\text{o.s.})} &= \overbrace{\sum_{\mathbf{r}\in\mathcal{S}_a} \hat{h}_V^{(\mathbf{r})}}^{\hat{H}_V^{\uparrow\downarrow}} + \overbrace{\sum_{\mathbf{r}\in\mathcal{S}_b} \hat{h}_V^{(\mathbf{r})}}^{\hat{H}_V^{\downarrow\uparrow}} \\ &= \sum_{\mathbf{r}\in\mathcal{S}} \hat{h}_V^{(\mathbf{r})} \end{aligned} \tag{1.10}$$

Sector	Term	Normal phase	Antiferromagnet	Superconductor
o.s.	Hartree			
	Fock	Prohibited	Prohibited	Prohibited
	Cooper	Prohibited	Prohibited	
s.s.	Hartree			
	Fock			
	Cooper	Prohibited	Prohibited	

Table 1.4 Summary of Wick's terms selection rules and relevant contributions to the effective hamiltonian for each phase limitedly to $\hat{H}_V$.

Here the notation of Fig. **??** is used. For each site i in a given sublattice, the nearest neighbors sites are four – all in the other sublattice. Similarly, the same-spin (s.s.) hamiltonian decomposes as

$$\hat{H}_V^{(\text{s.s.})} = -V \sum_{\mathbf{r}\in\mathcal{S}_a} \sum_{\boldsymbol{\delta},\sigma} (\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma} + \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma}) \quad \text{with} \quad \boldsymbol{\delta} \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}$$

Opposite-spin terms. Consider first the o.s. terms of Eq. (**??**): take e.g. the NNs term $\hat{n}_{i\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}$. As in Eq. (1.6), taking Wick's contractions,

$$\begin{aligned}\langle \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\rangle &= \langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle \\ &\simeq \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\rangle\langle \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Cooper}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\rangle}_{\text{Hartree}}\end{aligned}$$

Identical decompositions are given for all others NNs. Of the three terms above:

- The Coooper term breaks $\mathrm{U}^{\mathrm{c}}(1)$ charge symmetry, allowing for supeconducting instabilities;
- The Fock term breaks the $\mathrm{U}^z(1)$ symmetry, because it accounts for a site hop *plus* spin flip process;
- The Hartree term can reduce both translational invariance and spin rotational symmetry.

As a general rule of thumb, throughout this text we will discard the o.s. Fock term eveywhere, preserving a robust $\mathrm{U}^z(1)$ symmetry. For the normal state as well as for the AF instability only the Hartree term is to be accounted; in superconducting instability also the Cooper term contributes. Note that for superconducting instabilities, due to superexchange mechanism (as explained in App. **??**) the o.s. term account for singlet pairing as well as zero-projection triplet pairing. Which channel is preferred, is a matter of thermodynamic advantage.

Same-spin terms. Consider then the same-spin terms: take e.g. the term $\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}$. As above,

$$\begin{aligned}\langle \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\rangle &= \langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle \\ &= \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\rangle\langle \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Cooper}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\rangle}_{\text{Hartree}}\end{aligned}$$

Identical consideration as in the above paragraph hold for each term. The only difference with the o.s. terms is given by the Fock term: since the spin-flip process is absent, now the Fock fluctuations actually contribute as an effective NN hopping term. In other words, this term does not break $\mathrm{U}^z(1)$ symmetry and thus is perfectly legitimate in each phase. Tab. 1.4 summarizes the selection rules hereby presented.

1.3.2 Reciprocal space transform of the model interactions

It is useful to derive analytically the reciprocal-space form of both the U and V interactions. By the sign $\overset{\mathcal{F}}{=}$, we indicate a passage where a Fourier-transform is performed.

Non-local attraction. Consider a generic bond, say, the one connecting NNs $\mathbf{r}$ and $\mathbf{r} \pm \boldsymbol{\delta}$. Fourier transform it according to the convention

$$\hat{c}_{\mathbf{r}\sigma} \overset{\mathcal{F}}{=} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}\in \mathrm{BZ}} e^{-i\mathbf{k}\cdot\mathbf{r}} \hat{c}_{\mathbf{k}\sigma}$$

Then:

$$\begin{aligned}
\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'} &= \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'}\hat{c}_{\mathbf{r}\sigma} \\
&\overset{\mathcal{F}}{=} \frac{1}{(L_x L_y)^2} \sum_{\mathbf{k}_1,\cdots,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} e^{\pm i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}} \hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma}
\end{aligned}$$

Then, the interaction at site $\mathbf{r}$, spin σ with its NNs at spin σ' – indicated as $(\mathbf{r}\sigma\sigma')$ – is given by

$$\begin{aligned}
(\mathbf{r}\sigma\sigma') &= -V \sum_{\boldsymbol{\delta}} \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'} \\
&\overset{\mathcal{F}}{=} -\frac{V}{(L_x L_y)^2} \sum_{\boldsymbol{\delta}} \sum_{\mathbf{k}_1,\cdots,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} \\
&\qquad\qquad \times \left(e^{i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}} + e^{-i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}}\right) \hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma} \\
&= -\frac{2V}{(L_x L_y)^2} \sum_{\boldsymbol{\delta}} \sum_{\mathbf{k}_1,\cdots,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} \cos\left[(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}\right] \hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma}
\end{aligned}$$

The full non-local interaction is given by summing over all sites of $\mathcal{S}_a$ (which is, half the sites of $\mathcal{S}$). This gives back momentum conservation,

$$\frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{S}_a} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} = \frac{1}{2}\delta_{\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4}$$

Let $\mathbf{k}_1 + \mathbf{k}_2 = \mathbf{k}_3 + \mathbf{k}_4 = \mathbf{K}$, and define $\mathbf{k}$, $\mathbf{k}'$ such that

$$\mathbf{k}_1 \equiv \mathbf{K}+\mathbf{k} \qquad \mathbf{k}_2 \equiv \mathbf{K}-\mathbf{k} \qquad \mathbf{k}_3 \equiv \mathbf{K}-\mathbf{k}' \qquad \mathbf{k}_4 \equiv \mathbf{K}+\mathbf{k}' \qquad \delta\mathbf{k} \equiv \mathbf{k}-\mathbf{k}'$$

Sums over these variables must be intended as over the Brillouin Zone (BZ). Hence:

$$\begin{aligned}
\hat{H}_V &= \sum_{\mathbf{r}\in\mathcal{S}_a} \sum_{\sigma\sigma'} (\mathbf{r}\sigma\sigma') \\
&\overset{\mathcal{F}}{=} -\frac{V}{L_x L_y} \sum_{\sigma\sigma'} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \cos(\delta\mathbf{k}\cdot\boldsymbol{\delta})\, \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \\
&= -\frac{V}{L_x L_y} \sum_{\sigma\sigma'} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos(\delta k_x) + \cos(\delta k_y)\right] \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}
\end{aligned} \tag{1.11}$$

which decomposes in s.s. and o.s. channels as:

$$\begin{aligned}
\hat{H}_V \overset{\mathcal{F}}{=} &\overbrace{-\frac{V}{L_x L_y} \sum_{\sigma} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos(\delta k_x) + \cos(\delta k_y)\right] \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}^{\text{s.s.}} \\
&\underbrace{-\frac{V}{L_x L_y} \sum_{\sigma} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos(\delta k_x) + \cos(\delta k_y)\right] \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}_{\text{o.s.}}
\end{aligned}$$

The o.s. part can be simplified: since

$$\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} = \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}$$

due to fermionic anti-commutation rules, and since the composite transform

$$\mathbf{k} \to -\mathbf{k} \qquad \mathbf{k}' \to -\mathbf{k}'$$

leaves the form factor $\cos(\delta k_x) + \cos(\delta k_y)$ invariant, we reduce the spin sum to two times the $\sigma = \uparrow$ part,

$$\hat{H}_V^{(\text{o.s.})} = -\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos(\delta k_x) + \cos(\delta k_y)\right] \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \tag{1.12}$$

whil the explicit s.s. sector is given immediately by:

$$\hat{H}_V^{(\text{s.s.})} = -\frac{V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos(\delta k_x) + \cos(\delta k_y)\right] \times \left[\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\uparrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\uparrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} + \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\downarrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\downarrow}\right] \tag{1.13}$$

The result here obtained will be used various times in next chapters. Note that different Wick contraction schemes lead to different results:

$$\overbracket{c^\dagger_{\mathbf{K}+\mathbf{k}\sigma} c^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}} c_{\mathbf{K}-\mathbf{k}'\sigma'} c_{\mathbf{K}+\mathbf{k}'\sigma} \qquad \text{Cooper contraction} \tag{1.14}$$

$$\overbracket{c^\dagger_{\mathbf{K}+\mathbf{k}\sigma} c^\dagger_{\mathbf{K}-\mathbf{k}\sigma'} c_{\mathbf{K}-\mathbf{k}'\sigma'}} c_{\mathbf{K}+\mathbf{k}'\sigma} \qquad \text{Fock contraction} \tag{1.15}$$

$$\overbracket{c^\dagger_{\mathbf{K}+\mathbf{k}\sigma} c^\dagger_{\mathbf{K}-\mathbf{k}\sigma'} c_{\mathbf{K}-\mathbf{k}'\sigma'} c_{\mathbf{K}+\mathbf{k}'\sigma}} \qquad \text{Hartree contraction} \tag{1.16}$$

which will be used explicitly later on.

Local repulsion. A somewhat identical argument can be carried out for the local interaction,

$$\hat{H}_U = U \sum_{\mathbf{r}\in\mathcal{S}} \hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}\downarrow}$$

The only difference here is made by the fact that the interaction is repulsive on-site, thus the final structure factor in the Fourier transform disappears as well as the minus sign,

$$\hat{H}_U \overset{\mathcal{F}}{=} \frac{U}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \tag{1.17}$$

When contracting these fermionic operators, identical considerations as for the non-local counterpart (1.11) hold. Note the similarity of this Fourier transform with the o.s. sector of the non-local attraction, given in Eq. (1.11). With this, we conclude the general introduction to the model and pass to the setup of the Hartree-Fock (HF) algorithm.

1.4 General approach MFT through iterative Hartree-Fock algorithms

This section is devoted to the explanation of the general iterative strategy we use to simulate the EHM within MFT. The simplicity of the algorithm is only possible thanks to the violent reduction of complexity MFT is able to reach. We use an iterative self-consistent scheme, thus solving a finite set of equation searching for a fixed point in parameters space for each phase.

1.4.1 Sketch of the Hartree-Fock iterative algorithm

The general sketch of a self-consistent Hartree-Fock algorithm in our context is simple enough. Start from a fermionic quartic hamiltonian,

$$\hat{H} = \sum_{\alpha\beta} A_{\alpha\beta} \hat{c}^\dagger_\alpha \hat{c}_\beta + \sum_{\alpha\beta\gamma\delta} B_{\alpha\beta\gamma\delta} \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta + \text{h.c.} \tag{1.18}$$

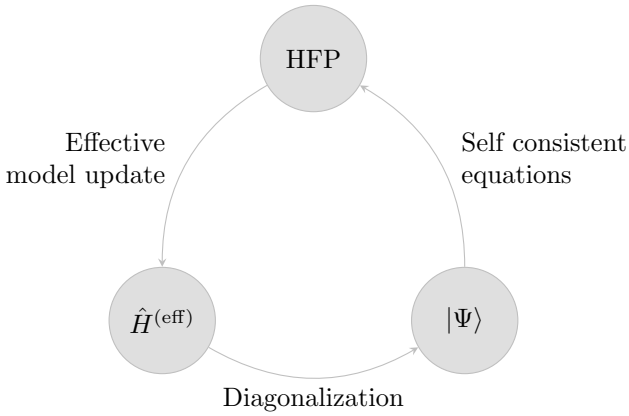


Figure 1.3 Schematization of the self-consistent Hartree Fock method described in Sec. 1.4. The effective model is solved, the solution is used to estimate self-consistently the Hartree Fock vector, which is then used to update iteratively the model itself. [Update including initializer and endpoint.]

where $A_{\alpha\beta}$, $B_{\alpha\beta\gamma\delta}$ are model parameters. We limit ourselves to quartic hamiltonians, but the idea can be extended. As discussed before, in this kind of context MFT is employed by applying Wick's theorem. This way, the quartic hamiltonian is effectively reduced to a quadratic hamiltonian,

$$\hat{H} \simeq \hat{H}^{(\mathrm{eff})} = \sum_{\alpha\beta} C_{\alpha\beta}\hat{c}^\dagger_\alpha \hat{c}_\beta + \sum_{\alpha\beta} D_{\alpha\beta}\hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta + \mathrm{h.c.} \tag{1.19}$$

where now the new parameters, which we will refer to as Hartree-Fock parameters (HFP), depend on the model itself via the equations

$$C_{\alpha\beta} = A_{\alpha\beta} + \sum_{\gamma\delta} \left[B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}\right] \left\langle \hat{c}^\dagger_\gamma \hat{c}_\delta \right\rangle$$
$$D_{\alpha\beta} = \sum_{\gamma\delta} B_{\alpha\beta\gamma\delta} \left\langle \hat{c}_\gamma \hat{c}_\delta \right\rangle$$

These equations are determined by many-body expectation values, which in general are computed by the means of thermal averages over the lowest energy density matrix. A self consistent Hartree-Fock scheme goes as follows:

1. Apply MFT to the quartic hamiltonian, recovering its general theoretic effective form;
2. Impose selection rules on the parameters $C_{\alpha\beta}$, $D_{\alpha\beta}$, eventually eliminating those not respecting the symmetry sector we are working in[1];
3. Choose a starting value and a coherent symmetry structure for the remaining parameters;
4. Enter iterative scheme of Fig. 1.3:
   a) Based on the current HFPs, determine $\mu[C, D]$ that gives the correct density;
   b) Diagonalize the effective hamiltonian $\hat{H}^{(\mathrm{eff})}[C, D] - \mu[C, D]\hat{N}$ and recover the ground-state $|\Psi[C, D]\rangle$;
   c) Compute the new HFPs $C'_{\alpha\beta}$, $D'_{\alpha\beta}$ via the self consistent equations;
   d) Initialize the new effective hamiltonian $\hat{H}^{(\mathrm{eff})}[C', D']$ and repeat the cycle until the HFPs have converged to stable values.

[1] As an example: if we choose not to break charge conservation, set $\forall\alpha, \beta: \ D_{\alpha\beta} = 0$.

The stable values we are referring to in the last point are our solutions. The resulting vector is a fixed point in the parametric recursive map defined by the set of MFT self-consistent equations.

Of course, to work any of these algorithms we need to know how to diagonalize the hamiltonian. In the context of the Extended Hubbad Model at thermal equilibrium, due to the phases we are simulating, we know analytically the diagonalized hamiltonian, and making use of this knowledge we are able to reduce the self consistent scheme to a simple iterative algorithm. An explicit example of HF algorithm used in present text, somewhat simplified due to the presence of a single HFP, is given in App. **??**.

1.4.2 Self-consistency and RMPs

As will be thoroughly discussed in the following derivations for each physical phase we treat, treating the EHM via MFT keeping under strict control the effects of Wick's theorem produces ground states whose parametric properties (band structures, gap functions, chemical potential) are controlled parametrically and self-consistently by the model two-body interactions $\hat{H}_U$ and $\hat{H}_V$. Let us define the concept of Renormalized model parameter (RMP), meaning one model that is effectively renormalized via interactions.

Let us invent a relevant example to explain the concept of RMP. Suppose that under MFT, for a given phase the decomposition $\hat{H}_V$ has a net effect of defining the RMP $\tilde{t}$,

$$t \xrightarrow{V} \tilde{t}$$

In other words, the decomposition of $\hat{H}_V$ produces a free Fermionic hamiltonian whose diffusive parameter is $\tilde{t}$. We define $\tilde{t}$ as an RMP, being the result of the original model parameter t plus the effect of interactions.

1.4.3 Fixed density simulations and the role of μ

In the context of the present text, all simulations are performed at fixed density, determining self-consistently the value of the chemical potential. This means that, for a given step i and the relative HFPs vector $\mathbf{v}_i$, and for a fixed density $n \in [0, 1]$, μ_i (the chemical potential at the current step) is determined as

$$\mu_i \quad : \quad \min_{\mu} \left\{ \frac{1}{2L_x L_y} \sum_{\mathbf{k}} \sum_{\sigma} \langle \hat{n}_{\mathbf{k}\sigma} \rangle \big|_{\mathbf{v}=\mathbf{v}_i} - n \right\} \tag{1.20}$$

being $\mathbf{v} = \mathbf{v}_i$ and indication that the averaging operation is performed over the ground state determined parametrically by the current value of the HFPs.

Hence, if we are able to perform averages and know both $\mathbf{v}_i$ and n, at step i we infer the correct chemical potential. Now, as we will discuss $\tilde{\mu}$ will turn out to be an RMP itself, meaning that theoretically

$$\mu \to \mu + F[\mathbf{v}] \equiv \tilde{\mu}[\mathbf{v}]$$

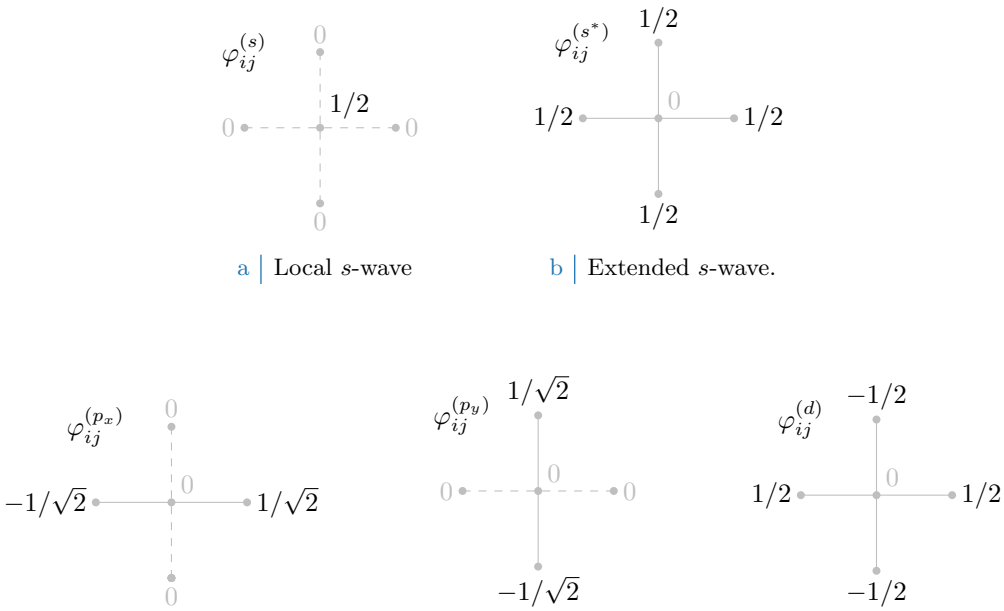
being F some functional of the HFPs. In other words, a common effect of interactions under MFT will be precisely to shift the chemical potential. At step i, keeping the density fixed, the chemical potential we are determining numerically and using in our equations is then $\tilde{\mu}_i$. In simpler words, the designed algorithm "does not know" about chemical potential renormalization and thus just determines the number that solves Eq. (1.20). But we know that this number is made of two contributions: the "physical" μ_i and the effective shift produced by the model itself, $F[\mathbf{v}_i]$. So, when it comes to μ_i, this consideration becomes important.

It is the case of the ground state energy estimation. Let us abstract from the EHM and MFT, and consider a generic hamiltonian $\hat{H}$. To find the ground state $|\Psi\rangle$ means precisely to solve

$$\frac{\delta}{\delta \langle \Psi |} \langle \Psi | \hat{H} | \Psi \rangle = 0$$

Now, working with a fixed particles number N means we are performing a constrained minimization (which is, taking the functional derivative) on the composite operator

$$\hat{H} - \mu(\hat{N} - N) \equiv \hat{K} + \mu N$$



a | Local s-wave b | Extended s-wave.

c | p_x-wave. d | p_y-wave. e | $d_{x^2-y^2}$-wave.

Figure 1.4 | Form factors at different topologies, as listed in Tab. 1.5. In figures five sites are represented: the hub site i and its four NN. Solid lines represent non-zero values for $\varphi_{\boldsymbol{\delta}}$, while dashed lines represent vanishing factors.

being μ the Lagrange multiplier, $\hat{N}$ the number operator and $\hat{K} \equiv \hat{H} - \mu\hat{N}$. Finding the constrained ground state parametrized by a vector $\mathbf{v}$ now means to solve

$$\mathbf{v} \implies |\Psi[\mathbf{v}]\rangle \quad : \quad \begin{cases} \dfrac{\delta}{\delta\langle\Psi|}\langle\Psi|\hat{K} + \mu N|\Psi\rangle\Big|_{|\Psi\rangle=|\Psi[\mathbf{v}]\rangle} = 0 \\ \langle\Psi[\mathbf{v}]|\hat{N}|\Psi[\mathbf{v}]\rangle = N \end{cases}$$

Of course, in the first line μN is derived to zero. The constraint is imposed in the second line. This means that, when it comes to determining the parametric ground state, $\hat{K}$ is functionally minimized. If in this context μ (the Lagrange multiplier) sums with other effective terms arisen from algebraic manipulations of $\hat{H}$ giving a grand total of $\tilde{\mu}$, the number operator $\hat{N}$ average is performed over the many-body ground state $|\Psi[\mathbf{v}]\rangle$ with the total effective chemical potential $\tilde{\mu}$, not μ. However, when it comes to computing energy,

$$\begin{aligned} \langle\hat{H}\rangle &= \langle\Psi[\mathbf{v}]|\hat{K}|\Psi[\mathbf{v}]\rangle + \mu N \\ &= \Omega[\mathbf{v}] + (\tilde{\mu}[\mathbf{v}] - F[\mathbf{v}])N \end{aligned}$$

being Ω the expectation value (EV) of $\hat{K}$. In the last passage we wrote everything in terms of functionals of the HFPs vector $\mathbf{v}$. Using this relation, when the algorithm has converged, we can estimate the physical energy.

1.5 Square lattice spatial symmetry channels

In next sections we introduce spatial symmetry structures and discuss their relation with point groups. We will aways refer to various spatial structures as "symmetry channels", throughout the system is able to establish a given order parameter defined in space.

Looking at $\hat{H}$, the main difference between its two-body interactions $\hat{H}_U$ and $\hat{H}_V$ of course relies in locality (first) and non-locality (second). Both operators in principle are able to break C_4 symmetry, as will later be discussed extensively.

Structure	Form factor	Graph
s-wave	$\varphi_{ij}^{(s)} = \delta_{ij}$	Fig. 1.4a
Extended s-wave	$\varphi_{ij}^{(s^*)} = \frac{1}{2}\left[\delta_{j=i+\delta_x} + \delta_{j=i-\delta_x} + \delta_{j=i+\delta_y} + \delta_{j=i-\delta_y}\right]$	Fig. 1.4b
p_x-wave	$\varphi_{ij}^{(p_x)} = \frac{1}{\sqrt{2}}\left[\delta_{j=i+\delta_x} - \delta_{j=i-\delta_x}\right]$	Fig. 1.4c
p_y-wave	$\varphi_{ij}^{(p_y)} = \frac{1}{\sqrt{2}}\left[\delta_{j=i+\delta_y} - \delta_{j=i-\delta_y}\right]$	Fig. 1.4d
$d_{x^2-y^2}$-wave	$\varphi_{ij}^{(d)} = \frac{1}{2}\left[\delta_{j=i+\delta_x} + \delta_{j=i-\delta_x} - \delta_{j=i+\delta_y} - \delta_{j=i-\delta_y}\right]$	Fig. 1.4e

Table 1.5 Form factors for a function defined on a square lattice, with a prefactor included for normalization. The last column indicates the graph representation of each contribution given in Fig. **??**. Subscript $x^2 - y^2$ is omitted for notational clarity.

Structure	Structure factor	Graph
s-wave	$\varphi_{\mathbf{k}}^{(s)} = 1$	Fig. 1.4a
Extended s-wave	$\varphi_{\mathbf{k}}^{(s^*)} = \cos k_x + \cos k_y$	Fig. 1.4b
p_x-wave	$\varphi_{\mathbf{k}}^{(p_x)} = i\sqrt{2}\sin k_x$	Fig. 1.4c
p_y-wave	$\varphi_{\mathbf{k}}^{(p_y)} = i\sqrt{2}\sin k_y$	Fig. 1.4d
$d_{x^2-y^2}$-wave	$\varphi_{\mathbf{k}}^{(d)} = \cos k_x - \cos k_y$	Fig. 1.4e

Table 1.6 Structure factors derived from the correlation structures of Tab. 1.5. The functions hereby defined are orthonormal, and are plotted in Fig- 1.5.

For the 2D square lattice, various spatial symmetries are possible. Let i be a specific site, and consider a generic complex function g_{ij} whose domain is given by the site itself and its four NNs (the domain concides with the points of Fig. 1.2). Then it is immediate to see that the function can be written in the following basis

$$g_{ij} = \sum_{\gamma} g^{(\gamma)} \varphi_{ij}^{(\gamma)}$$

where $g^{(\gamma)} \in \mathbb{C}$ are the g_{ij} symmetries-decomposition coefficients while $\varphi_{ij}^{(\gamma)}$ are the form factors listed in Tab. 1.5 and plotted in Fig. 1.5, a simple orthonormal rearrangement of the harmonics basis.

When dealing with phases which respect translational symmetry, or at most reduce it to a weaker symmetry by shrinking the Brillouin Zone, it' useful to work in reciprocal space, as discussed in Sec. **??**. As said, a particular phase can show specific spatial symmetries, and so will the Hartree Fock parameters. The key point to notice is that self-consistent equations involve sums (or integrals) to be computed over fermionic states, which can be heavily simplified by employing said symmetries. Consider the form factors of Tab. 1.5, and their Fourier transforms of Tab. 1.6. They satisfy the relations:

$$\frac{1}{L_x L_y} \sum_{ij} \varphi_{ij}^{(\gamma)} \varphi_{ij}^{(\gamma)*} = \delta_{\gamma\gamma'} \qquad \frac{1}{L_x L_y} \sum_{\mathbf{k}} \varphi_{\mathbf{k}}^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*} = \delta_{\gamma\gamma'}$$

If these symmetries are developed by the HFPs, it's of no use to integrate over the entire BZ, since different regions linked by periodicity of the form factors will lead to redundant calculations.

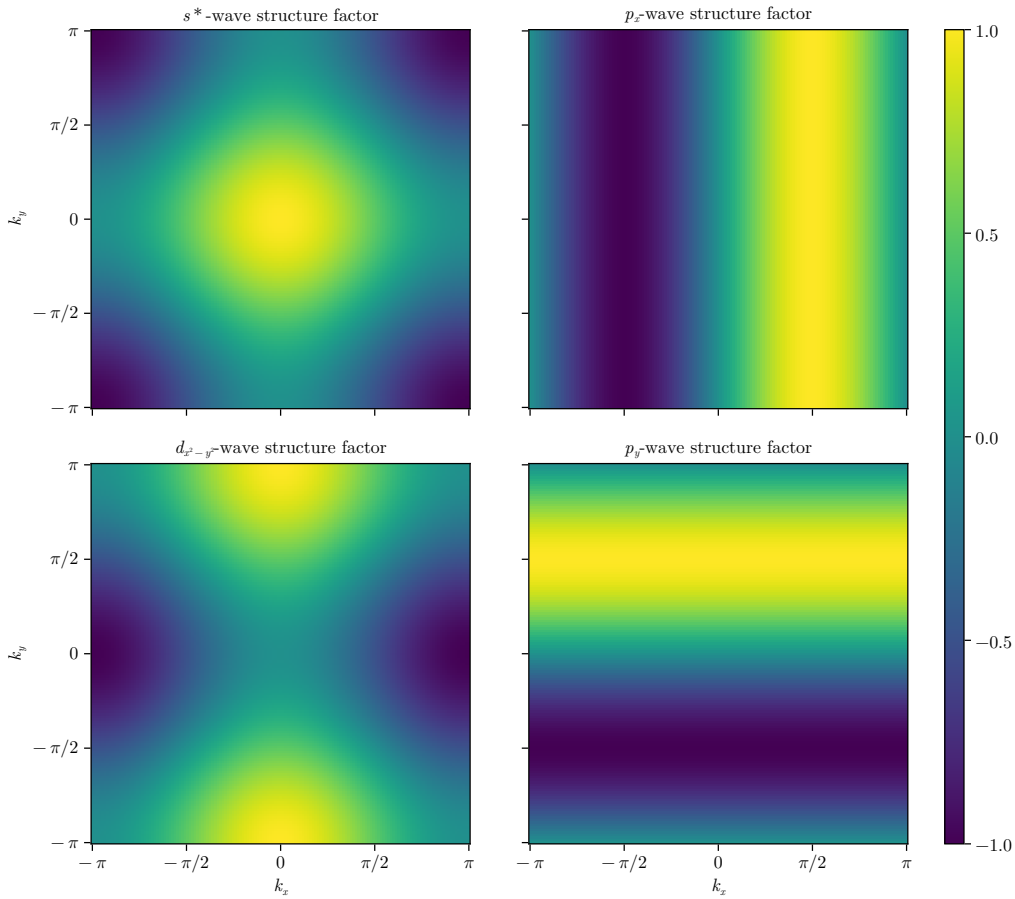


Figure 1.5 | Representation of the various structure factors listed in Tab. 1.6 on the BZ. For the s^* and $d_{x^2-y^2}$-wave factors, the real part is plotted; for the p_ℓ factors, the imaginary part is plotted. [Rotate d-wave]

Finally, a very useful detail is given by the following argument. Consider the equation

$$\cos(\delta k_x) + \cos(\delta k_y) = \cos k_x \cos k'_x + \sin k_x \sin k'_x + \cos k_y \cos k'_y + \sin k_y \sin k'_y$$

For the sake of readability, the notations

$$c_\ell \equiv \cos k_\ell \qquad s_\ell \equiv \sin k_\ell \qquad c'_\ell \equiv \cos k'_\ell \qquad s'_\ell \equiv \sin k'_\ell$$

are used. Group the four terms above,

$$\underbrace{\left(c_x c'_x + c_y c'_y\right)}_{\text{Symmetric}} + \underbrace{\left(s_x s'_x + s_y s'_y\right)}_{\text{Anti-symmetric}} \tag{1.21}$$

The first two exhibit inversion symmetry for both arguments $\mathbf{k}$, $\mathbf{k}'$; the second two exhibit anti-symmetry. Decoupling the symmetric part, we obtain

$$c_x c'_x + c_y c'_y = \frac{1}{2}(c_x + c_y)(c'_x + c'_y) + \frac{1}{2}(c_x - c_y)(c'_x - c'_y)$$

which finally gives:

$$\begin{aligned}
\cos(\delta k_x) + \cos(\delta k_y) = \frac{1}{2}(c_x + c_y)(c'_x + c'_y) & \qquad (s^*\text{-wave}) \\
+ s_x s'_x & \qquad (p_x\text{-wave}) \\
+ s_y s'_y & \qquad (p_y\text{-wave}) \\
+ \frac{1}{2}(c_x - c_y)(c'_x - c'_y) & \qquad (d_{x^2-y^2}\text{-wave})
\end{aligned}$$

or, in a more compact form

$$\cos(\delta k_x) + \cos(\delta k_y) = \frac{1}{2}\sum_{\gamma} \varphi_{\mathbf{k}}^{(\gamma)} \varphi_{\mathbf{k}'}^{(\gamma)*} \quad \text{for } \gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\} \tag{1.22}$$

$$= \frac{1}{2}\sum_{\gamma} \varphi_{\mathbf{k}}^{(\gamma)*} \varphi_{\mathbf{k}'}^{(\gamma)} \tag{1.23}$$

This decomposition will become particularly handy in later calculations.